Cyclist Project Report

Abstract

This project analyzes Cyclistic's 2025 bike-share trip data to identify behavioral differences between annual members and casual riders using Python, MySQL, and Power BI. The insights generated from ride patterns, duration, and usage trends help support data-driven marketing strategies aimed at converting casual riders into annual members.

Introduction

Cyclistic operates a bike-sharing service that offers both annual memberships and casual ride options to its customers. This project analyzes 12 months of trip data to understand the differences in riding behavior between annual members and casual riders, providing insights that can support targeted marketing strategies and increase membership conversions.

Problem Statement

Cyclistic aims to increase the number of annual memberships, as annual members are more profitable than casual riders. The challenge is to understand how annual members and casual riders use the bike-sharing service differently and identify behavioral patterns that can help develop effective marketing strategies to convert casual riders into annual members..

Objectives

1. To analyze 12 months of Cyclistic bike-share trip data and understand customer riding behavior.

2. To compare ride patterns between annual members and casual riders based on ride frequency and duration.

3. To identify trends in bike usage across different days, months, and hours of the year.

4. To uncover key behavioral differences that influence membership and casual ride usage.

5. To provide data-driven insights and recommendations that can help Cyclistic convert casual riders into annual members.

6. Build an interactive Power BI dashboard.

Dataset Description

The Cyclistic bike-share trip data for all 12 months of 2025 was downloaded and imported into Python using the Pandas and Glob libraries. The individual monthly datasets were combined into a single dataframe to create a consolidated dataset for analysis.

Tools & Technologies

- Python (Pandas, NumPy) – Data Cleaning

- SQL – Business Query Analysis

- Power BI – Dashboard Visualization

- Jupyter Notebook – Development Environment

Data Collection and Preparation

- Imported all 12 monthly Cyclistic trip datasets using the Pandas and Glob libraries.
- Used Glob to automatically locate and load multiple CSV files from a folder.
- Combined the monthly datasets into a single dataframe to create a unified dataset for analysis.

- This ensured that all trips from the entire year were available for comprehensive analysis.

	ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual
0	7569BC890583FCD7	classic_bike	2025-01-21 17:23:54.538	2025-01-21 17:37:52.015	Wacker Dr & Washington St	KA1503000072	McClurg Ct & Ohio St	TA1306000029	41.883143	-87.637242	41.892592	-87.617289	member
1	013609308856B7FC	electric_bike	2025-01-11 15:44:06.795	2025-01-11 15:49:11.139	Halsted St & Wrightwood Ave	TA1309000061	Racine Ave & Belmont Ave	TA1308000019	41.929147	-87.649153	41.939743	-87.658865	member
2	EACACD3CE0607C0D	classic_bike	2025-01-02 15:16:27.730	2025-01-02 15:28:03.230	Southport Ave & Waveland Ave	13235	Broadway & Cornelia Ave	13278	41.948226	-87.664071	41.945529	-87.646439	member
3	EAA2485BA64710D3	classic_bike	2025-01-23 08:49:05.814	2025-01-23 08:52:40.047	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member
4	7F8BE2471C7F746B	electric_bike	2025-01-16 08:38:32.338	2025-01-16 08:41:06.767	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member

- Used merged_df.describe() to analyze statistical summaries such as mean, median, minimum, maximum, and standard deviation.

```python
merged_df.describe(include='all')
```

Python

	ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual
count	5552994	5552994	5552994	5552994	4368321	4368321	4309689	4309689	5.552994e+06	5.552994e+06	5.547459e+06	5.547459e+06	5552994
unique	5552994	2	5551938	5550528	1893	3360	1901	3383	NaN	NaN	NaN	NaN	2
top	7569BC890583FCD7	electric_bike	2025-01-29 17:03:06.348	2025-08-18 19:30:47.144	Kingsbury St & Kinzie St	CHI01747	Kingsbury St & Kinzie St	CHI01747	NaN	NaN	NaN	NaN	member
freq	1	3604953	2	23	40612	54075	40013	53592	NaN	NaN	NaN	NaN	3553497
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.190366e+01	-8.764648e+01	4.190411e+01	-8.764683e+01	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.443323e-02	2.726130e-02	4.460469e-02	2.746640e-02	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.164850e+01	-8.789000e+01	4.149000e+01	-8.810000e+01	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.188213e+01	-8.766000e+01	4.188241e+01	-8.766000e+01	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.189859e+01	-8.764177e+01	4.189964e+01	-8.764275e+01	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.193000e+01	-8.762991e+01	4.193000e+01	-8.763000e+01	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.207000e+01	-8.752000e+01	4.221000e+01	-8.742000e+01	NaN

- Checked for missing/null values using df.isnull().sum() And

```
[7] merged_df.isnull().sum()

ride_id                     0
rideable_type               0
started_at                  0
ended_at                    0
start_station_name    1184673
start_station_id      1184673
end_station_name      1243305
end_station_id        1243305
start_lat                   0
start_lng                   0
end_lat                  5535
end_lng                  5535
member_casual               0
dtype: int64
```

- Replaced missing values in station name and station ID columns with "Unknown".
- This approach preserved ride records while clearly identifying unavailable station information.

- Standardized column names by converting them to lowercase and replacing spaces with underscores for better consistency and easier querying.

- Prevented unnecessary data loss during the cleaning process.

```
station_cols = ['start_station_name', 'end_station_name', 'start_station_id', 'end_station_id']
for col in station_cols:
    merged_df[col] = merged_df[col].fillna('Unknown')
print("Missing values in station columns have been filled with 'Unknown' the columns are:")
                                                                                          Python

Missing values in station columns have been filled with 'Unknown' the columns are:
```

- Removed records containing missing ending latitude and longitude values.
- Geographic coordinates are important location attributes and cannot be accurately replaced.
- Removing incomplete records improved data reliability.

```
[18] merged_df = merged_df.dropna(subset=['end_lat', 'end_lng'])

[19] merged_df.isnull().sum()

ride_id               0
rideable_type         0
started_at            0
ended_at              0
start_station_name    0
start_station_id      0
end_station_name      0
end_station_id        0
start_lat             0
start_lng             0
end_lat               0
end_lng               0
member_casual         0
ride_length           0
day_of_week           0
dtype: int64
```

- Converted the started_at and ended_at columns into datetime format.
- Enabled accurate time-based calculations and feature extraction.
- Datetime conversion is required for calculating trip durations and temporal analysis.

```
merged_df['started_at'] = pd.to_datetime(merged_df['started_at'])
merged_df['ended_at'] = pd.to_datetime(merged_df['ended_at'])
```

- Created a new column called ride_length.
- Calculated ride duration by subtracting trip start time from trip end time.
- This metric serves as one of the most important variables in the analysis.

	ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual	ride_length
0	7569BC890583FCD7	classic_bike	2025-01-21 17:23:54.538	2025-01-21 17:37:52.015	Wacker Dr & Washington St	KA1503000072	McClurg Ct & Ohio St	TA1306000029	41.883143	-87.637242	41.892592	-87.617289	member	0 days 00:13:57.477000
1	013609308856B7FC	electric_bike	2025-01-11 15:44:06.795	2025-01-11 15:49:11.139	Halsted St & Wrightwood Ave	TA1309000061	Racine Ave & Belmont Ave	TA1308000019	41.929147	-87.649153	41.939743	-87.658865	member	0 days 00:05:04.344000
2	EACACD3CE0607C0D	classic_bike	2025-01-02 15:16:27.730	2025-01-02 15:28:03.230	Southport Ave & Waveland Ave	13235	Broadway & Cornelia Ave	13278	41.948226	-87.664071	41.945529	-87.646439	member	0 days 00:11:35.500000
3	EAA2485BA64710D3	classic_bike	2025-01-23 08:49:05.814	2025-01-23 08:52:40.047	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	0 days 00:03:34.233000
4	7F8BE2471C7F746B	electric_bike	2025-01-16 08:38:32.338	2025-01-16 08:41:06.767	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	0 days 00:02:34.429000

- Extracted the day of the week from the ride start timestamp.
- Enabled analysis of riding behavior across weekdays and weekends.
- Helped identify commuting and leisure usage patterns.

	ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual	ride_length	day_of_week
0	7569BC890583FCD7	classic_bike	2025-01-21 17:23:54.538	2025-01-21 17:37:52.015	Wacker Dr & Washington St	KA1503000072	McClurg Ct & Ohio St	TA1306000029	41.883143	-87.637242	41.892592	-87.617289	member	0 days 00:13:57.477000	3
1	013609308856B7FC	electric_bike	2025-01-11 15:44:06.795	2025-01-11 15:49:11.139	Halsted St & Wrightwood Ave	TA1309000061	Racine Ave & Belmont Ave	TA1308000019	41.929147	-87.649153	41.939743	-87.658865	member	0 days 00:05:04.344000	0
2	EACACD3CE0607C0D	classic_bike	2025-01-02 15:16:27.730	2025-01-02 15:28:03.230	Southport Ave & Waveland Ave	13235	Broadway & Cornelia Ave	13278	41.948226	-87.664071	41.945529	-87.646439	member	0 days 00:11:35.500000	5
3	EAA2485BA64710D3	classic_bike	2025-01-23 08:49:05.814	2025-01-23 08:52:40.047	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	0 days 00:03:34.233000	5
4	7F8BE2471C7F746B	electric_bike	2025-01-16 08:38:32.338	2025-01-16 08:41:06.767	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	0 days 00:02:34.429000	5

- Removed rides lasting less than one minute.
- Such records are likely caused by accidental unlocks, test rides, or system errors.
- Removing unrealistic trips improved analytical accuracy.

```
merged_df= merged_df[merged_df['ride_length']>pd.Timedelta(minutes=1)]
merged_df.head()
```

Python

	ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual	ride_length	day_of_week
0	7569BC890583FCD7	classic_bike	2025-01-21 17:23:54.538	2025-01-21 17:37:52.015	Wacker Dr & Washington St	KA1503000072	McClurg Ct & Ohio St	TA1306000029	41.883143	-87.637242	41.892592	-87.617289	member	0 days 00:13:57.477000	3
1	013609308856B7FC	electric_bike	2025-01-11 15:44:06.795	2025-01-11 15:49:11.139	Halsted St & Wrightwood Ave	TA1309000061	Racine Ave & Belmont Ave	TA1308000019	41.929147	-87.649153	41.939743	-87.658865	member	0 days 00:05:04.344000	0
2	EACACD3CE0607C0D	classic_bike	2025-01-02 15:16:27.730	2025-01-02 15:28:03.230	Southport Ave & Waveland Ave	13235	Broadway & Cornelia Ave	13278	41.948226	-87.664071	41.945529	-87.646439	member	0 days 00:11:35.500000	5
3	EAA2485BA64710D3	classic_bike	2025-01-23 08:49:05.814	2025-01-23 08:52:40.047	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	0 days 00:03:34.233000	5
4	7F8BE2471C7F746B	electric_bike	2025-01-16 08:38:32.338	2025-01-16 08:41:06.767	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	0 days 00:02:34.429000	5

- Checked for duplicate rides using the unique ride_id field.
- Removed duplicate entries to prevent double-counting trips.

- Ensured each ride was represented only once in the dataset.

```python
merged_df =merged_df.drop_duplicates(subset=['ride_id'])
print(f"Total duplicate rows removed: {len(merged_df) - len(merged_df.drop_duplicates(subset=['ride_id']))}")
```

Python

```
Total duplicate rows removed: 0
```

- Converted ride duration into seconds and minutes.
- These numerical measures simplify aggregation and statistical calculations.
- Ride duration metrics were later used for user behavior comparison.
- Extracted month and hour from ride start timestamps.
- Enabled monthly trend analysis and hourly usage pattern analysis.
- These features help identify seasonal and daily riding behavior.

	ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual	day_of_week	ride_length_in_seconds	ride_length_minutes	month	hour
0	7569BC890583FCD7	classic_bike	2025-01-21 17:23:54.538	2025-01-21 17:37:52.015	Wacker Dr & Washington St	KA1503000072	McClurg Ct & Ohio St	TA1306000029	41.883143	-87.637242	41.892592	-87.617289	member	3	837.477	13.957950	January	17
1	013609308856B7FC	electric_bike	2025-01-11 15:44:06.795	2025-01-11 15:49:11.139	Halsted St & Wrightwood Ave	TA1309000061	Racine Ave & Belmont Ave	TA1308000019	41.929147	-87.649153	41.939743	-87.658865	member	0	304.344	5.072400	January	15
2	EACACD3CE0607C0D	classic_bike	2025-01-02 15:16:27.730	2025-01-02 15:28:03.230	Southport Ave & Waveland Ave	13235	Broadway & Cornelia Ave	13278	41.948226	-87.664071	41.945529	-87.646439	member	5	695.500	11.591667	January	15
3	EAA2485BA64710D3	classic_bike	2025-01-23 08:49:05.814	2025-01-23 08:52:40.047	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	5	214.233	3.570550	January	8
4	7F8BE2471C7F746B	electric_bike	2025-01-16 08:38:32.338	2025-01-16 08:41:06.767	Southport Ave & Waveland Ave	13235	Southport Ave & Roscoe St	13071	41.948226	-87.664071	41.943739	-87.664020	member	5	154.429	2.573817	January	8

- Exported the cleaned dataset into a MySQL database using SQLAlchemy.
- Stored data in a structured format for efficient querying.
- Enabled SQL-based business analysis.

SQL Analysis

Business Questions

- Which user type uses bikes more?

member_casual	Total_rides
member	3484216
casual	1915835

- Which user type rides longer?

member_casual	avg_rides_in_type
casual	19.98129637714991
member	12.200936479252851

- What days do users ride most?

member_casual	day_of_week	totat_ride
casual	6	306472
casual	5	247896
casual	4	212748
casual	3	216936
casual	2	219239
casual	1	316966
casual	0	395578
member	6	518546
member	5	565180
member	4	540214
member	3	552541
member	2	493322
member	1	374538
member	0	439875

- What hours are busiest?

member_casual	hour	total_rides
casual	17	183273
casual	16	168794
casual	18	156926
casual	15	148057
casual	14	133241
casual	13	126739
casual	12	123712
casual	19	116403
casual	11	105314
casual	20	83797
casual	10	82454
casual	21	71203

- Which months have highest usage?

member_casual	month	total_rides
casual	August	323545
casual	July	308443
casual	June	278699
casual	September	254728
casual	October	214390
casual	May	175667
casual	April	105271
casual	November	94698

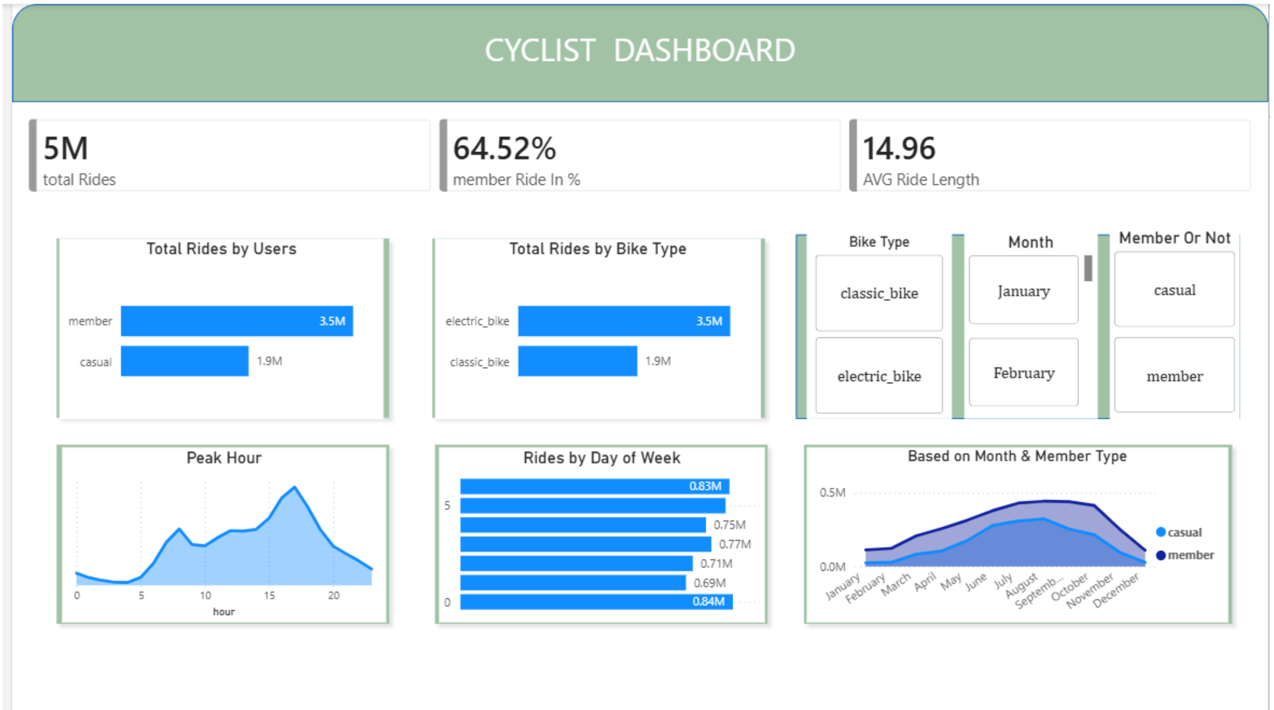
- Which bike types are preferred?

rideable_type	total_rides
electric_bike	3457557
classic_bike	1942494

- Average ride duration by weekday

member_casual	avg_time	day_of_week
member	11.643174972356473	4
member	11.747963997133624	5
member	11.782422299363809	2
member	11.88126054512978	3
member	12.156485563800933	6
member	13.314678402992783	0
member	13.465322176832794	1
casual	16.438572269618	4

Dashboard Development



The Power BI dashboard includes:

- Dashboard Overview

- An interactive Power BI dashboard was developed to visualize key performance indicators and riding behavior patterns of Cyclistic users.
- The dashboard enables comparison between annual members and casual riders through various metrics and trend analyses.

- Key Performance Indicators (KPIs)

- Total rides recorded during the analysis period reached approximately 5 million trips.
- Annual members accounted for 64.52% of total rides, indicating higher engagement compared to casual riders.
- The average ride duration was approximately 14.96 minutes.

- User Type Analysis

- Annual members generated significantly more rides than casual riders.
- This suggests that members use the bike-sharing service more frequently, likely for regular commuting and daily transportation needs.

- Bike Type Analysis

- Electric bikes were the most frequently used bike type among riders.
- The popularity of electric bikes indicates a preference for convenience and reduced riding effort.

Peak Hour Analysis

- Ride activity increased during daytime hours and peaked during the late afternoon and evening period.
- This pattern suggests that many users utilize the service for commuting and daily travel purposes.

- Day of Week Analysis

- Ride activity varied across the week, with certain days experiencing higher usage than others.
- Understanding these patterns helps identify customer behavior during weekdays and weekends.

- Monthly Trend Analysis

- Both annual members and casual riders showed increased activity during warmer months.
- Ride volume peaked during the summer season and declined toward the winter months.
- Annual members consistently recorded higher ride counts throughout the year compared to casual riders.

- Dashboard Filters

- Interactive filters were implemented for bike type, month, and rider category.
- These filters allow users to explore specific segments and perform detailed analysis based on business requirements.

- Business Value

- The dashboard provides a clear understanding of customer usage patterns.
- The insights generated can support marketing decisions aimed at increasing annual memberships and improving customer engagement.

Key Findings

- Annual members contribute the majority of rides
- Electric bikes are the most preferred bike type.
- Ride activity peaks during afternoon and evening hours.
- Usage increases significantly during summer months.
- Members consistently use the service more frequently than casual riders.

Recommendations

- Offer targeted membership discounts to frequent casual riders to increase annual subscriptions.

- Launch seasonal marketing campaigns during high-demand summer months.

- Promote membership benefits through personalized email and app notifications.

- Increase customer engagement using social media and digital marketing channels.

- Focus promotional efforts during peak riding hours and high-usage periods.

Conclusion

The Cyclistic bike-share analysis successfully identified key differences between annual members and casual riders by examining ride frequency, duration, bike preferences, and usage patterns across different times of the year. The findings indicate that annual members use the service more consistently, while casual riders show stronger seasonal and recreational usage patterns. These insights can help Cyclistic develop targeted marketing strategies to convert casual riders into annual members and support long-term business growth.